# Supplementary material for the work: "Path Integral Monte Carlo on a Sphere"

Riccardo Fantoni*
*Università di Trieste, Dipartimento di Fisica, strada Costiera 11, 34151 Grignano (Trieste), Italy*
(Dated: March 8, 2026)

Supplementary material to the article "Path Integral Monte Carlo on a Sphere".

Keywords: Path Integral; Monte Carlo; Sphere; Quantum Fluid; Bosons; Fermions; Anyons; Thermodynamics; Structure; Superfluidity; Sign Problem

## I. INTRODUCTION

We here report the FORTRAN code listing of the code used in the simulations developed in the publication "Path Integral Monte Carlo on a Sphere". This is shown in Appendix A. The model studied in that publication was that of a many body fluid of $N$ particles at $\boldsymbol{r}_i$, $i = 1, 2, \ldots, N$, on a sphere. Where $\boldsymbol{r} = (\theta, \varphi)$ with $\theta \in ]-\pi/2, \pi/2]$ the polar angle and $\varphi \in ]-\pi, \pi]$ the azimuthal angle.

The plane waves [1] Path Integral Monte Carlo (PIMC) method used in our computer experiments is the one described in Ref. [2]. A single MC step is made of the exchange of a randomly chosen pair of particles through the *Lévy construction* of two Brownian bridges between two randomly chosen timeslices on different particles paths (see Section V.G of Ref. [2] and references therein for the multislice move) and of a displacement of all the timeslices of a particle path chosen at random (the singleslice move). The exchange move allows to sample the permutation sum necessary in the calculation of the symmetric Bose-Einstein density matrix. The extent of the singleslice displacement is chosen at the beginning of the run through the input variable `ALPHA` or it can be adjusted during the run every `IRATIO` MC steps so to have acceptance ratios closest to 1/2 (this functionality is commented out in the present listing).

We discretize the imaginary time between 0 and `BETA` into `FTNO` timeslices of length $\tau$=`LS`=`BETA/FTNO`. One usually wants to compare simulations at fixed `LS`. The multislice move requires the reconstruction of a randomly chosen number `MBM`≤`MBMM`≤`FNTO` of timeslices between a randomly chosen timeslice `CP` on a randomly chosen particle `IP` and a timeslice on a randomly chosen particle `KP`. Here `MBMM` is the maximum number of timeslices in a Brownian bridge. The singleslice move requires the displacement of all the `FTNO` timeslices of the path of the particle `IP` chosen at random. So a single MC `STEP` requires a maximum of 2`MBMM`+`FTNO` timeslice moves. In the code we wrote the FORTRAN instructions for the multislice move in lowercase and the ones for the singleslice move in uppercase. The simulation sets `KP`=`IP` in the excahnge move if the logical variable `IFB` is set to `FALSE`. So that if it is set to `TRUE` Bose-Einstein statistics is used, otherwise is used Boltzmann statistics. The simulation uses only the multislice move if the logical variable `IFBRIDGE` is `TRUE` and only the singleslice move if the logical variable `IFDISP` is `TRUE`. The simulation lasts for a total number of MC steps `NSTEP`. All this variables can be chosen in the `data-gr-rs.in` simulation configuration input file at startup.

On a sphere it is necessary to choose `DIM`= 2 with a number of particles $N$=`NP` of mass `FTM` with natural units $\hbar = k_B = 1$. One has to specify also the radius of the sphere `RADIUS` and the absolute temperature `TEMP`.

The pair potential between the particles `POTK` is encoded in `SUBROUTINE POT` at the end of the code listing. It includes the Coulomb potential that can be chosen by setting the string `POTK` equal to `COULOMB`. The ideal gas is obtained choosing the string `FREE`. This can be set up in the `data-gr-rs.in` simulation configuration input file. The variables `RCUT`, `SIG`, `EPSR`, `EPSA`, and `EPS` in the simulation configuration input file are used to specify other properties of a non Coulomb pair potential.

The simulation calculates both thermodynamic quantities and static structural properties. For the thermodynamic quantities we measure: the kinetic energy `KE`, the physical potential energy `V`, the geometric effective potential energy `W`$= -\sum_{i=1}^{N} \ln \sqrt{g(\boldsymbol{r}_i)}/\tau$ where $g$ is the determinant of the sphere metric tensor, and the superfluid fraction `SF`. These quantities are stored in the output file `fort.9` every `IPRINT` MC steps. Whereas the path configurations output file `CNFILE` is stored in a *.xyz* format every `ISAVE` MC steps. The first `IEQUI` MC steps are used for the equilibration of the Markov chain. Quantities are only calculated after these equilibration steps. For the structural properties we measure the radial distribution function $g(r)$, with $r$ the Euclidean distance, which is written in the output file `rdf.dat`. In the current implementation of the code, the $g(r)$ is only calculated during the displacement move every `ICALCG` timeslices.

If the variable `INIT` is set to 0 the simulation starts from a path with all particles at all timeslices on the equator $\theta = 0$ at $\varphi = 0$, if it is set to 1 it starts from a path with the particles randomly distributed uniformly on the equator with all timeslices copied above the first one for each particle, if it is set to 2 it reads the paths configuration from the file `CNFILE` stored on disk, if it is

* riccardo.fantoni@scuola.istruzione.it
[1] For a coherent states PIMC see Ref. [1].

set to 3 it does a complete restart from the file `CNFILE` and the frozen configuration `back.dat` both stored on disk.

The variable `SEED`, set in the simulation configuration input file, is the positive integer seed of the pseudo random number generator.

For the RPIMC algorithm also used in the work for the simulations of fermions [3, 4] and anyons we modified the code presented here as described in the main text of the article.

## ACKNOWLEDGMENTS

I would like to thank prof. Saverio Moroni for his support in the development of the brownian bridge and the consequent particles permutation sampling.

## Appendix A: The code

This is the code used for the PIMC computer experiment. We list here the main FORTRAN code `pimc-sph-rs.f` with the included `mc-sph.par` parameters file. And the input simulation configuration file `data-gr-rs.in` that is read at the beginning of the run.

```
*****************************************************************************
*** pimc-sph-rs.f ***********************************************************
*****************************************************************************

      PROGRAM PIMC
      IMPLICIT NONE
C     ***************************************************************
C     ** MONTE CARLO SIMULATION PROGRAM FOR [PW] PATH INTEGRAL      **
C     **                                                             **
C     ** on a sphere of radius RADIUS                               **
C     **                                                             **
C     ** space dimensions:    DIM=2                                 **
C     ** number of particles: NP                                    **
C     ** number of timesteps: FTN0                                  **
C     ** units:               hbar=k_B=1                            **
C     ** BETA=1/TEMP=FTN0*LS  (LS imaginary time spacing)           **
C     **                                                             **
C     ** OBSERVABLES:                                               **
C     ** kinetic energy       KE                                    **
C     ** potential energy     V                                     **
C     ** forces virial        W                                     **
C     ** superfluid fraction  SF                                    **
C     ** radial distribution function on 'rdf.dat'                 **
C     ***************************************************************
      INCLUDE       'mc-sph.par'                        ! parameters

      INTEGER*4     SEED
      INTEGER*8     IDIM, STEP, I, J, K
      INTEGER*8     C1, CP, IP, KP, NIP, NKP, LL, OUT1, OUT2, PIP
      INTEGER*8     INIT, NSTEP, NS, IPRINT, ISAVE, IRATIO, IEQUI
      INTEGER*8     MBMM, MCM, PREV(MNP)
      INTEGER*8     CYC(MNP), NSWAP

      REAL*8        DENS, TEMP, BETA, DENSLJ
      REAL*8        DRMAX, ALPHA, CDIM
      REAL*8        RANF, DUMMY, SR9, SR3
      REAL*8        VLRC, VLRC6, VLRC12, WLRC, WLRC6, WLRC12
      REAL*8        ACM, ACATMA, ACATMAS, ACATMAB
      REAL*8        V, VNEW, VOLD, VEND, DELTV, VN, VS
      REAL*8        W, WNEW, WOLD, WEND, DELTW, WN, WS
      REAL*8        KE, KENEW, KEOLD, DELTKE, KEEND
      REAL*8        AVV, ACV, ACVSQ, FLV
      REAL*8        AVW, ACW, ACWSQ, FLW
      REAL*8        AVKE, ACKE, ACKESQ, FLKE
      REAL*8        ACT, ACTNEW, ACTOLD, DELTACT, DELTACTB
      REAL*8        RXIOLD(MDIM), RXINEW(MDIM)
      REAL*8        RXP(MDIM,0:N), RXPP(MDIM,0:N)
      REAL*8        STD, PS, KKK, VVV, WWW
      REAL*8        RATIO, RATIOS, RATIOB
      REAL*8        WIND, ACWIND, SFI, SF, IC
      REAL*8        GA

      INTEGER*8     BIN, MAXBIN, MAXBIN2, HIST(1000000), ICALCG
      REAL*8        CONST, RLOWER, RUPPER
      REAL*8        TSTEP, NIDEAL, GR

      CHARACTER     CNFILE*30, POTK*10

      LOGICAL       OVRLAP, IFB, IFDISP, IFBRIDGE

C       PI = ACOS(-1.d0)
      OPEN(UNIT = 16, FILE = 'back.dat', STATUS ='UNKNOWN')

C     ***************************************************************
C     ** READ INPUT DATA                                            **
C     ***************************************************************

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)'')
      WRITE(*,'('' PLANE WAVES                                    '')'')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM              '')'')
      OPEN (UNIT=10, FILE='data-gr-rs.in', STATUS='UNKNOWN')
      READ (10,*) I
      READ (10,*) DIM
      READ (10,*) I
      READ (10,*) SEED
      READ (10,*) I
      READ (10,*) IFB
      READ (10,*) I
      READ (10,*) NP
      READ (10,*) I
      READ (10,'(A)') POTK
      READ (10,*) I
      READ (10,*) FTN0
      READ (10,*) I
      READ (10,*) FTM
      READ (10,*) I
      READ (10,*) NSTEP
      READ (10,*) I
      READ (10,*) IPRINT
      READ (10,*) I
      READ (10,*) ISAVE
      READ (10,*) I
      READ (10,*) IEQUI
      READ (10,*) I
      READ (10,*) IRATIO
      READ (10,*) I
      READ (10,*) ICALCG
      READ (10,*) I
      READ (10,'(A)') CNFILE
      READ (10,*) I
      READ (10,*) INIT
      READ (10,*) I
      READ (10,*) RADIUS
      READ (10,*) I
      READ (10,*) TEMP
      READ (10,*) I
      READ (10,*) ALPHA
      READ (10,*) I
      READ (10,*) MBMM
      READ (10,*) I
      READ (10,*) RCUT
      READ (10,*) I
      READ (10,*) SIG
      READ (10,*) I
      READ (10,*) EPSR
      READ (10,*) I
      READ (10,*) EPSA
      READ (10,*) I
      READ (10,*) EPS
      READ (10,*) I
      READ (10,*) IFDISP
      READ (10,*) I
      READ (10,*) IFBRIDGE
      CLOSE (UNIT=10)
      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)'') DIM
      WRITE(*,'('' ************************************      ''/)'')

      GOTO 1111                     ! comment if input from keyboard

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)'')
      WRITE(*,'('' PLANE WAVES                                    '')'')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM              '')'')

      WRITE(*,'('' ENTER THE SPATIAL DIMENSIONS                   '')'')
      READ (*,*) DIM
      WRITE(*,'('' ENTER SEED FOR RANDOM SEQUENCE                 '')'')
      READ (*,*) SEED
      WRITE(*,'('' IF BOSE (T/F)                                  '')'')
      READ (*,*) IFB
      WRITE(*,'('' ENTER THE NUMBER OF PARTICLES < MNP            '')'')
      READ (*,*) NP
      WRITE(*,'('' ENTER TYPE OF PAIR POTENTIAL (Many Body)       '')'')
      READ (*,'(A)') POTK
      WRITE(*,'('' ENTER THE NUMBER OF DISCRETIZATIONS < N        '')'')
      READ (*,*) FTN0
      WRITE(*,'('' ENTER THE BARE MASS                            '')'')
      READ (*,*) FTM
      WRITE(*,'('' ENTER NUMBER OF CYCLES                         '')'')
      READ (*,*) NSTEP
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN OUTPUT LINES ''')'')
      READ (*,*) IPRINT
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN DATA SAVES       '')'')
      READ (*,*) ISAVE
      WRITE(*,'('' ENTER NUMBER OF STEPS FOR EQUILIBRATION        '')'')
      READ (*,*) IEQUI
      WRITE(*,'('' ENTER INTERVAL FOR UPDATE OF MAX. DISPL.       '')'')
      READ (*,*) IRATIO
      WRITE(*,'('' ENTER INTERVAL FOR RDF CALCULATION             '')'')
      READ (*,*) ICALCG
      WRITE(*,'('' ENTER THE CONFIGURATION FILE NAME              '')'')
      READ (*,'(A)') CNFILE
      WRITE(*,'('' ENTER 0 INIT Z, 1 INIT, 2 REST C, 2 REST F ''')'')
      READ (*,*) INIT
      WRITE(*,'('' ENTER THE SPHERE RADIUS                        '')'')
      READ (*,*) RADIUS
      WRITE(*,'('' ENTER THE TEMPERATURE                          '')'')
      READ (*,*) TEMP
      WRITE(*,'('' ENTER MAX. DISPLACEMENT DRMAX/SIGMA            '')'')
      READ (*,*) ALPHA
      WRITE(*,'('' ENTER MAXIMUM NUMBER OF BRIDGE TIMESLICES      ''')'')
      READ (*,*) MBMM
```

```
      WRITE(*,'(’’ ENTER THE POTENTIAL CUTOFF DISTANCE (~ 2.5)’’)’)
      READ (*,*) RCUT
      WRITE(*,'(’’ ENTER SIG                                  ’’)’)
      READ (*,*) SIG
      WRITE(*,'(’’ ENTER EPSR                                 ’’)’)
      READ (*,*) EPSR
      WRITE(*,'(’’ ENTER EPSA                                 ’’)’)
      READ (*,*) EPSA
      WRITE(*,'(’’ ENTER EPS                                  ’’)’)
      READ (*,*) EPS
      WRITE(*,'(’’ IF DISPLACEMENT MOVE (T/F)                 ’’)’)
      READ (*,*) IFDISP
      WRITE(*,'(’’ IF BRIDGE MOVE (T/F)                       ’’)’)
      READ (*,*) IFBRIDGE
      WRITE(*,'(’’ IN DIMENSION (1,2,3,...)’’I12/)’) DIM
      WRITE(*,'(’’ ************************************      ’’/)’)

 1111 CONTINUE

      IF (MBMM .GT. FTN0) THEN
         WRITE(*,*) "number of timeslices in bridge > ",FTN0
         STOP
      ENDIF

      IF (DIM .NE. 2) THEN
         WRITE(*,*) "DIM not equal to 2"
         STOP
      ENDIF

C     ** INITIALIZE RANDOM NUMBER GENERATOR **

      CALL SRAND ( SEED )
      IF (SEED.EQ.1) THEN
         CALL SRAND ( 0 )
      ENDIF

C     ** DENSITY **

      DENS = NP/4.d0/PI/RADIUS**2

C     ** INVERSE TEMPERATURE BETA **

      BETA = (1.d0/TEMP)

C     ** IMAGINARY TIMESTEP **

      LS  = BETA/FTN0      ! time step
      STD = (LS/FTM)**0.5  ! standatd deviation of free-particle

C     ** MOMENT OF INERTIA OF A UNITARY SPHERICAL SHELL **

      IC = 2*(NP*FTM)/3

C    ** WRITE INPUT DATA **

      WRITE(*,'(’’ SEED                      ’’,I10   )’) SEED
      WRITE(*,'(’’ POTENTIAL                 ’’,A     )’) POTK
      WRITE(*,'(’’ DIMENSIONS                ’’,I10   )’) DIM
      WRITE(*,'(’’ NUMBER OF PARTICLES       ’’,I10   )’) NP
      WRITE(*,'(’’ NUMBER OF CYCLES          ’’,I10   )’) NSTEP
      WRITE(*,'(’’ NUMBER OF EQUIL. STEPS    ’’,I10   )’) IEQUI
      WRITE(*,'(’’ OUTPUT FREQUENCY          ’’,I10   )’) IPRINT
      WRITE(*,'(’’ SAVE FREQUENCY            ’’,I10   )’) ISAVE
      WRITE(*,'(’’ RATIO UPDATE FREQUENCY    ’’,I10   )’) IRATIO
      WRITE(*,'(’’ CONFIGURATION FILE  NAME  ’’,A     )’) CNFILE
      WRITE(*,'(’’ TEMPERATURE               ’’,E10.4 )’) TEMP
      WRITE(*,'(’’ DENSITY                   ’’,E10.4 )’) DENS
      WRITE(*,'(’’ PARTICLE MASS             ’’,E10.4 )’) FTM
      WRITE(*,'(’’ # TIME SLICES             ’’,I10   )’) FTN0
      WRITE(*,'(’’ TIME STEP                 ’’,E10.4 )’) LS

C    ** CONVERT INPUT DATA TO PROGRAM UNITS **

C       SIGMA  = ( DBLE ( NP ) / DENS ) ** ( 1.0 / DIM )
      SIGMA   = 2*PI
      SIGM(1) = PI
      SIGM(2) = 2*PI
      DRMAX   = ALPHA * SIGMA

      IF (POTK .EQ. ’LJ’) THEN
         DENSLJ = DENS * SIG ** DBLE( DIM )
C         RCUT = SIGMA/2.d0/SIG
      ENDIF

      MAXBIN  = INT ( 2*RADIUS/DELR )     ! Euclidean
C      MAXBIN  = INT (PI*RADIUS/DELR )     ! Geodesic
      MAXBIN2 = INT ( MAXBIN/2.0 )
      DO I = 1, MAXBIN
         HIST(I) = 0
      ENDDO

      IF (DIM .eq. 3) CONST = 4.d0 * PI * DENS / 3.d0
      IF (DIM .eq. 2) CONST = PI * DENS
      IF (DIM .eq. 1) CONST = 2.d0 * DENS

C     ** INITIALIZE CONFIGURATION **

      DO I = 1, NP
         NEXT(I) = I
         LEXT(I) = I
         DO J = 1, FTN0
            MEXT(J,I)  = I
         ENDDO
      ENDDO

C     ** ZERO ACCUMULATORS **
      ACV     = 0.0
      ACVSQ   = 0.0
      FLV     = 0.0
      ACW     = 0.0
      ACWSQ   = 0.0
      FLW     = 0.0
      ACKE    = 0.0
      ACKESQ  = 0.0
      FLKE    = 0.0
      ACWIND  = 0.0
      ACM     = 0.0
      ACATMA  = 0.0
      ACATMAB = 0.0
      ACATMAS = 0.0

      IF ( INIT .EQ. 0 ) THEN
         PRINT*,"PATHS INITIALIZATION ..............."
         RX=0.                ! all paths zeroed
         NS = NSTEP
      ELSEIF ( INIT .EQ. 1) THEN
         PRINT*,"PATHS INITIALIZATION ..............."
C     ** READ INITIAL CONFIGURATION **
         CALL INITCNSPH ( CNFILE ) ! random configuration on equator
         CALL READCN ( CNFILE )
         NS = NSTEP
      ELSEIF ( INIT .EQ. 2) THEN
         PRINT*,"RESTART FROM CONFIG  ..............."
C     ** READ INITIAL CONFIGURATION **
         CALL READCN ( CNFILE )
         NS = NSTEP
      ELSEIF ( INIT .EQ. 3) THEN
         PRINT*,"RESTART FULL         ..............."
         CALL READCN ( CNFILE )
         READ(16,*)(HIST(I),I=1,MAXBIN),
     &          ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWIND,ACM,NS
         NS=NS+NSTEP
      ELSE
         WRITE (*,*) ’INIT different from 0,1,2,3’
         STOP
      ENDIF

C     ** CALCULATE LONG RANGE CORRECTIONS    **
C     ** SPECIFIC TO THE LENNARD JONES FLUID **
      IF (DIM .EQ. 1) CDIM = 1
      IF (DIM .EQ. 2) CDIM = PI
      IF (DIM .EQ. 3) CDIM = 2*PI

      SR3 = - RCUT ** (- 6. + DIM)/(-6. + DIM)
      SR9 = - RCUT ** (- 12. + DIM)/(-12. + DIM)

      VLRC12 =   4 * EPS * CDIM * DENSLJ * NP * SR9
      VLRC6  = - 4 * EPS * CDIM * DENSLJ * NP * SR3
      VLRC   =   VLRC12 + VLRC6
      WLRC12 =   4.0  * VLRC12
      WLRC6  =   2.0  * VLRC6
      WLRC   =   WLRC12 + WLRC6

C     ** WRITE OUT SOME USEFUL INFORMATION **

      WRITE(*,'(’’ SIGMA                     ’’,E10.4)’) SIGMA
      WRITE(*,'(’’ MAXIMUM DISPLACEMENT      ’’,E10.4)’) DRMAX

C    ** CALCULATE INITIAL ENERGY AND CHECK FOR OVERLAPS **

      CALL SUMUP (POTK, OVRLAP, KE, V, W)

      IF (POTK .EQ. ’LJ’) THEN
         VS = ( V + VLRC )
         WS = ( W + WLRC )
      ELSE
         VS = V
         WS = W
      ENDIF

      WRITE(*,'(’’ INITIAL V                 ’’,E10.4)’) VS
      WRITE(*,'(’’ INITIAL W                 ’’,E10.4)’) WS
      WRITE(*,'(’’ INITIAL KE                ’’,E10.4)’) KE

      WRITE(*,'(//’’ START OF MARKOV CHAIN               ’’//)’)
      WRITE(*,'(’’  NMOVE     RATIO       ACTION          ’’/)’)

C     ****************************************************************
C     ** LOOPS OVER ALL CYCLES AND ALL TIME SLICES                   **
C     ****************************************************************

      DO 100 STEP = NS-NSTEP+1, NS

         CP = INT(FTN0*RANF(DUMMY))+1    ! select a random timeslice
         MBM = INT((MBMM-1)*RANF(DUMMY))+2  ! # timeslices in bridge
         IP = INT(NP*RANF(DUMMY))+1      ! select a particle
         KP = INT(NP*RANF(DUMMY))+1      ! select another particle
         IF (.NOT.IFB) KP = IP           ! for boltzmann statistics
         IF (IFDISP) GOTO 77   ! uncomment if only displacement move

C     ****************************************************************
C     ** BRIDGE&SWAP MOVE (BOSE STATISTICS)                          **
C     ****************************************************************

C     direct mapping from sphere rx to plane rp
         call mappingd()
         call equiv(rpn,rp)

         mcm = cp+mbm-floor((cp+mbm-.1)/ftn0)*ftn0

         nip=next(ip)
         nkp=next(kp)

         vold=0.d0
         wold=0.d0
         if(cp+mbm.le.ftn0)then
            do i=cp+1,mcm-1
               call pppnergy ( potk, rx(:,i,ip), ip,
     :                 rx(:,i,kp), kp, i,
     :                 vvv, www )
               vold=vold+vvv
               wold=wold+www
```

```
            enddo
         else
            do i=cp+1,ftn0
               call pppnergy ( potk, rx(:,i,ip), ip,
     :                rx(:,i,kp), kp, i,
     :                vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
            do i=1,mcm-1
               call pppnergy ( potk, rx(:,i,nip), nip,
     :                rx(:,i,nkp), nkp, i,
     :                vvv, www )
               vold=vold+vvv
               wold=wold+www
            enddo
         endif

         keold=0.d0
         do k=1,np
            if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
               do i=1,ftn0
                  call kknergy ( rx(:,i,k), k, i,
     :                kkk )
                  keold=keold+kkk
               enddo
            endif
         enddo

c     bridge move
         if(cp+mbm.le.ftn0)then
            call bridge(rp(:,cp,ip),rp(:,mcm,kp),std,
     :           rxp,out1)
            if (ip.ne.kp) then
               call bridge(rp(:,cp,kp),rp(:,mcm,ip),std,
     :                rxpp,out2)
            endif
         else
            call bridge(rp(:,cp,ip),rp(:,mcm,nkp),std,
     :           rxp,out1)
            if (ip.ne.kp) then
               call bridge(rp(:,cp,kp),rp(:,mcm,nip),std,
     :                rxpp,out2)
            endif
         endif

c     mapping rxp and rxpp from plane to sphere

         vnew=0.d0
         wnew=0.d0
         ll=0
         if(cp+mbm.le.ftn0)then
            do i=cp+1,mcm-1
               ll=ll+1
               if (ip.eq.kp) then
                  call ppnergy ( potk, rxp(:,ll), ip,
     :                rxp(:,ll), kp, i,
     :                vvv, www )
               else
                  call ppnergy ( potk, rxp(:,ll), ip,
     :                rxpp(:,ll), kp, i,
     :                vvv, www )
               endif
               vnew=vnew+vvv
               wnew=wnew+www
            enddo
         else
            do i=cp+1,ftn0
               ll=ll+1
               if (ip.eq.kp) then
                  call ppnergy ( potk, rxp(:,ll), ip,
     :                rxp(:,ll), kp, i,
     :                vvv, www )
               else
                  call ppnergy ( potk, rxp(:,ll), ip,
     :                rxpp(:,ll), kp, i,
     :                vvv, www )
               endif
               vnew=vnew+vvv
               wnew=wnew+www
            enddo
            do i=1,mcm-1
               ll=ll+1
               if (ip.eq.kp) then
                  call ppnergy ( potk, rxp(:,ll), nip,
     :                rxp(:,ll), nkp, i,
     :                vvv, www )
               else
                  call ppnergy ( potk, rxpp(:,ll), nip,
     :                rxp(:,ll), nkp, i,
     :                vvv, www )
               endif
               vnew=vnew+vvv
               wnew=wnew+www
            enddo
         endif
         deltv=vnew-vold

c     execute the bridge move
         acatmab = acatmab + 1.d0
         call updaten(cp,rxp,ip,nkp)
         if (ip.ne.kp) then
            call updaten(cp,rxpp,kp,nip)
            call swapn(cp,ip,nip,kp,nkp)
            call switch(next(ip),next(kp))
            call switch(mext(cp,ip),mext(cp,kp))
            acatmas = acatmas + 1.d0
         endif

c     calculate the ratio of transition probabilities
         call gauss(cp,ga)

c     inverse mapping
         call equiv(rp,rpn)
         call equiv(rpn,rx)
         call mappingi()

         kenew=0.d0
         do k=1,np
            if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
               do i=1,ftn0
                  call kknergy ( rx(:,i,k), k, i,
     :                kkk )
                  kenew=kenew+kkk
               enddo
            endif
         enddo
         deltke=kenew-keold

c     calculate the acceptance probability
         call acc_p(deltv,deltke,ps)
         ps=ps*ga

c     accept/reject
         if (ps.gt.ranf(dummy)) then
            ke=ke+deltke
            v=v+vnew-vold
            w=w+wnew-wold
         else
            call equiv(rx,rpn)
            acatmab = acatmab - 1.d0
            if (ip.ne.kp) then
               call switch(next(ip),next(kp))
               call switch(mext(cp,ip),mext(cp,kp))
               acatmas = acatmas - 1.d0
            endif
         endif

c     build the permutations cycles in lext
         call permcyc()
c     count number of swaps of two particles
         call cycles(cyc,nswap)

c     balance ke between bridge and displace
c          call sumupke ( ke )

         IF (STEP.GT.IEQUI) THEN
            ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

            IF (POTK .EQ. 'LJ') THEN
               VN = ( V + VLRC )
            ELSE
               VN = V - W
               WN = W
            ENDIF
            CALL CWIND( WIND )
            IF (WIND.GT.3*BETA*IC*2/FTM**2) WIND=3*BETA*IC*2/FTM**2

C     ** ACCUMULATE AVERAGES **

            ACV    = ACV    + VN
            ACVSQ  = ACVSQ  + VN*VN
            ACW    = ACW    + WN
            ACWSQ  = ACWSQ  + WN*WN
            ACKE   = ACKE   + KE
            ACKESQ = ACKESQ + KE*KE
            ACWIND = ACWIND + WIND
         ENDIF

         IF (IFBRIDGE) GOTO 97

C     ****************************************************************
C     ** ENDS BRIDGE&SWAP MOVE                                      **
C     ****************************************************************

 77      CONTINUE

C     ****************************************************************
C     ** DISPLACEMENT MOVE                                          **
C     ****************************************************************

C     ** PREVIOUS IP **

         DO I=1,NP
            J=NEXT(I)
            PREV(J)=I
         ENDDO

         NIP = NEXT(IP)
         PIP = PREV(IP)

         DO 90 C1 = 1, FTN0

            DO IDIM = 1, DIM
               RXIOLD(IDIM) = RX(IDIM,C1,IP)
            ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE OLD CONFIGURATION **

            CALL PENERGY ( POTK, RXIOLD, IP, C1,
     :              VOLD, WOLD )

            CALL KENERGY ( RXIOLD, IP, C1,
     :              KEOLD )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

            ACTOLD = KEOLD + VOLD

C     ** MOVE I AND PICKUP THE CENTRAL IMAGE **
```

```
            DO IDIM = 1, DIM
               RXINEW(IDIM) = RXIOLD(IDIM) +
     :              ( 2.0 * RANF ( DUMMY ) - 1.0 )*DRMAX
               RXINEW(IDIM) = RXINEW(IDIM) -
     :                DNINT ( RXINEW(IDIM)/SIGM(IDIM) )*SIGM(IDIM)
            ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE NEW CONFIGURATION **

            CALL PENERGY ( POTK, RXINEW, IP, C1,
     :            VNEW, WNEW )

            CALL KENERGY ( RXINEW, IP, C1,
     :            KENEW )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

            ACTNEW = KENEW + VNEW

            DELTV  = VNEW  - VOLD
            DELTW  = WNEW  - WOLD
            DELTKE = KENEW - KEOLD

C     ** CHECK FOR ACCEPTANCE **

            DELTACT  = ACTNEW - ACTOLD
            DELTACTB = LS*DELTACT

            IF ( EXP ( - DELTACTB ) .GT. RANF ( DUMMY ) ) THEN
               V          = V  + DELTV
               W          = W  + DELTW
               KE         = KE + DELTKE
               ACATMA = ACATMA + 1.0

               DO IDIM = 1, DIM
                  RX(IDIM,C1,IP)  = RXINEW(IDIM)
C     IMAGINARY TIME PERIODIC BOUNDARY CONDITIONS
                  IF (C1.EQ.1) THEN
                     RX(IDIM,FTNO+1,PIP)=RX(IDIM,C1,IP)
                  ENDIF
                  IF (C1.EQ.FTNO) THEN
                     RX(IDIM,0,NIP)=RX(IDIM,C1,IP)
                  ENDIF
               ENDDO
            ENDIF

            IF (STEP.GT.IEQUI) THEN
               ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

C     HISTOGRAM FOR g(r)
               IF ( MOD ( C1, ICALCG ) .EQ. 0 ) THEN
                  CALL GOFR( C1, MAXBIN, HIST )
               ENDIF

               IF (POTK .EQ. 'LJ') THEN
                  VN = ( V + VLRC )
               ELSE
                  VN = V - W
                  WN = W
               ENDIF
               CALL CWIND( WIND )
               IF (WIND.GT.3*BETA*IC*2/FTM**2) WIND=3*BETA*IC*2/FTM**2

C     ** ACCUMULATE AVERAGES **

               ACV    = ACV    + VN
               ACVSQ  = ACVSQ  + VN*VN
               ACW    = ACW    + WN
               ACWSQ  = ACWSQ  + WN*WN
               ACKE   = ACKE   + KE
               ACKESQ = ACKESQ + KE*KE
               ACWIND = ACWIND + WIND
            ENDIF

C     ***************************************************************
C     ** END DISPLACEMENT MOVE                                     **
C     ***************************************************************

 90         CONTINUE

C     ***************************************************************
C     ** ENDS LOOP OVER TIME SLICES                                **
C     ***************************************************************

 97         CONTINUE

C     ** WRITE OUT THE INSTANTANEOUS VALUES ON FORT.9 **

          SFI = WIND  *FTM**2/BETA/IC/2
          SF  = ACWIND*FTM**2/BETA/IC/2/ACM

          CALL FLUSH(9)
          IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN
             WRITE(9,*) STEP, DIM*NP/2/LS-KE/FTNO, VN/FTNO, WN/FTNO, SFI
          ENDIF

C     ** CREATE POSITION DISTRIBUTION **

C          CALL DISTR(30_8,STEP*FTNO*NP/30)

C     ** PERFORM PERIODIC OPERATIONS **

          IF ( MOD ( STEP, IRATIO ) .EQ. 0 ) THEN

C     ** ADJUST MAXIMUM DISPLACEMENT **

            RATIO  = ACATMA  / DBLE ( FTNO * IRATIO )
            RATIOB = ACATMAB / DBLE ( IRATIO )
            RATIOS = ACATMAS / DBLE ( IRATIO )

            IF ( RATIO .GT. 0.5 ) THEN

C               DRMAX  = DRMAX  * 1.05

            ELSE

C               DRMAX  = DRMAX  * 0.95

            ENDIF

            ACATMA  = 0.0
            ACATMAB = 0.0
            ACATMAS = 0.0

          ENDIF

          IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN

C     ** WRITE OUT RUNTIME INFORMATION **

            WRITE(*,*) '          step            acm  dr/sig    ard
     :     arb       ars'
            WRITE(*,'(2(2XI13),(2Xf6.4),4(2Xf8.6))') STEP, INT(ACM),
     :                                ALPHA, RATIO, RATIOB, RATIOS
            WRITE(*,*) '            ke              v              w
     :        sf'
            WRITE(*,'(4(2XE13.7))') KE/FTNO, VN/FTNO, WN/FTNO,
     :                                SFI
            WRITE(*,*) '          <ke>            <v>            <w>
     :         <sf>'
            WRITE(*,'(4(2XE13.7))') ACKE/ACM/FTNO, ACV/ACM/FTNO,
     :                              ACW/ACM/FTNO, SF
            WRITE(*,*) '          nswap'
            WRITE(*,'(I3)') NSWAP
          ENDIF

          IF ( MOD ( STEP, ISAVE ) .EQ. 0 ) THEN

C     ** WRITE OUT THE CONFIGURATION AT INTERVALS **

            CALL WRITCN ( CNFILE )
            REWIND(16)
            CALL FLUSH(16)
            WRITE(16,*)(HIST(I),I=1,MAXBIN),
     &            ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWIND,ACM,STEP

C         ** WRITE OUT THE g(r) **

            IF ( STEP .GT. IEQUI ) THEN
               TSTEP = DBLE((STEP - IEQUI)*INT(FTNO/ICALCG))
               OPEN (UNIT = 10, FILE = 'rdf.dat', STATUS = 'UNKNOWN')
               DO BIN = 1, MAXBIN
                  RLOWER = DBLE ( BIN - .5)*DELR
                  RUPPER = RLOWER + DELR
                  NIDEAL = (RUPPER/2/RADIUS)**DIM-
     :                 (RLOWER/2/RADIUS)**DIM
                  GR = DBLE(HIST(BIN))/TSTEP/DBLE(NP)**2/NIDEAL
                  CALL FLUSH(10)
                  WRITE (10,*) DBLE(BIN-1)*DELR, GR
               ENDDO
               CLOSE (UNIT = 10)
            ENDIF

          ENDIF

 100    CONTINUE

C     ***************************************************************
C     ** ENDS THE LOOP OVER CYCLES                                 **
C     ***************************************************************

        WRITE(*,'(//'' END OF MARKOV CHAIN          '')//)')

C     ** CHECKS FINAL VALUE OF THE POTENTIAL ENERGY IS CONSISTENT **

        CALL SUMUP (POTK, OVRLAP, KEEND, VEND, WEND)

        IF ( ABS(VEND - V) .GT. 1.0d-03 ) THEN

           WRITE(*,'(''  PROBLEM WITH V ENERGY !!!'')')
           WRITE(*,'('' VEND   = '', E20.6)') VEND
           WRITE(*,'('' V      = '', E20.6)') V

        ENDIF

        IF ( ABS(KEEND - KE) .GT. 1.0d-03 ) THEN

           WRITE(*,'(''  PROBLEM WITH KE ENERGY !!!'')')
           WRITE(*,'('' KEEND  = '', E20.6)') KEEND
           WRITE(*,'('' KE     = '', E20.6)') KE

        ENDIF

C     ** WRITE OUT THE FINAL CONFIGURATION FROM THE RUN **

        CALL WRITCN ( CNFILE )

C     ** CALCULATE AND WRITE OUT RUNNING AVERAGES **

        AVV   = ACV / ACM
        ACVSQ = ( ACVSQ / ACM ) - AVV ** 2
        AVKE   = ACKE / ACM
        ACKESQ = ( ACKESQ / ACM ) - AVKE ** 2

C    ** CALCULATE FLUCTUATIONS **

        IF ( ACVSQ  .GT. 0.0 ) FLV  = SQRT ( ACVSQ/ACM )/FTNO
        IF ( ACKESQ .GT. 0.0 ) FLKE = SQRT ( ACKESQ/ACM )/FTNO
```

```
      WRITE(*,'(/'' AVERAGES ''/ )')
      WRITE(*,'('' <V/N>                 = '',E12.6)') AVV/FTN0
      WRITE(*,'('' <KE/N>                = '',E12.6)') AVKE/FTN0

      WRITE(*,'(/'' FLUCTUATIONS ''/)')

      WRITE(*,'('' FLUCTUATION IN <V/N>  = '',E12.6)') FLV
      WRITE(*,'('' FLUCTUATION IN <KE/N> = '',E12.6)') FLKE
      WRITE(*,'(/'' END OF SIMULATION '')')

C     ******************************************************************
C     ******************************************************************
C     ******************************************************************
C     ** THE END                                                      **
C     ******************************************************************
C     ******************************************************************
C     ******************************************************************
      CLOSE(UNIT = 16)
      STOP
      END


      subroutine permcyc()
      implicit none
!     given a particle i, lext(i) returns the cycle it belongs to
      INCLUDE       'mc-sph.par'

      integer*8 i,j,k,ii(mnp),jj,kk,ll

      kk=1
      do i=1,np
         ll=1
         ii(ll)=i
 1       jj=next(ii(ll))
         do j=1,ll
            if(jj.eq.ii(j))goto 2
         enddo
         ll=ll+1
         ii(ll)=jj
         goto 1
 2       do k=1,ll
            lext(ii(k))=kk
         enddo
         kk=kk+1
      enddo

      return
      end


      subroutine cycles(cyc,ns)
      implicit none
!     cyc(i) returns the number of particles in cycle of type i
!     ns returns the total number of 2 particles swaps
      INCLUDE       'mc-sph.par'
      integer*8 i,cyc(mnp),ns

      ns=0
      cyc=0
      do i=1,np
         cyc(lext(i))=cyc(lext(i))+1
      enddo
      do i=1,np
         if (cyc(i).ne.0) ns=ns+cyc(i)-1
      enddo

      return
      end


      subroutine switch(i,j)
!     switch i and j
      implicit none
      integer*8 i,j,k
      k=i
      i=j
      j=k
      return
      end


      subroutine acc_p(dv,dke,p)
      implicit none
!     complete acceptance probability
      INCLUDE       'mc-sph.par'

      real*8 p,dke,dv

      p=exp(-ls*dv)        ! contribution from the pair potential
      p=p*exp(-ls*dke)     ! contribution from the kinetic energy

      return
      end


      subroutine updaten(j,rxp,ip,nip)
      implicit none
!     updates a portion of the current path x using the proposed path xp
      INCLUDE       'mc-sph.par'

      integer*8 ip,nip,kp,nkp,j,l,k,idim
      integer*8 mcm
      real*8 rxp(mdim,0:n)

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      l=0
      if(j+mbm.le.ftn0)then
         do k=j+1,mcm-1
            l=l+1
            do idim=1,dim
               rpn(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
      else
         do k=j+1,ftn0
            l=l+1
            do idim=1,dim
               rpn(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
         do k=1,mcm-1
            l=l+1
            do idim=1,dim
               rpn(idim,k,nip)=rxp(idim,l)
            enddo
         enddo
      endif
      do idim=1,dim
         rpn(idim,ftn0+1,ip)=rpn(idim,1,nip)
         rpn(idim,0,nip)=rpn(idim,ftn0,ip)
      enddo

      return
      end


      subroutine swapn(j,ip,nip,kp,nkp)
      implicit none
!     swap the final part of the paths
      INCLUDE       'mc-sph.par'

      integer*8 ip,nip,kp,nkp,j,k,idim
      integer*8 mcm
      real*8    rr

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      if(j+mbm.le.ftn0)then
         do k=mcm,ftn0
            do idim=1,dim
               rr=rpn(idim,k,ip)
               rpn(idim,k,ip)=rpn(idim,k,kp)
               rpn(idim,k,kp)=rr
            enddo
         enddo
         do idim=1,dim
            rpn(idim,ftn0+1,ip)=rpn(idim,1,nkp)
            rpn(idim,ftn0+1,kp)=rpn(idim,1,nip)
            rpn(idim,0,nip)=rpn(idim,ftn0,kp)
            rpn(idim,0,nkp)=rpn(idim,ftn0,ip)
         enddo
      endif
      return
      end


      subroutine bridge(x0,x1,std,xnew,out)
      implicit none
!     sample m gaussians with std from xnew(0)=x0 to xnew(ftn0)=x1
      INCLUDE       'mc-sph.par'

      integer*8 l1,l2,l3,j,idim,out
      real*8 std,d,s,xi
      real*8 x0(mdim),x1(mdim),xnew(mdim,0:n)

      l3=mbm
      do idim=1,dim
         xnew(idim,0)=x0(idim)
         xnew(idim,l3)=x0(idim)+(x1(idim)-x0(idim))-
     :          dnint((x1(idim)-x0(idim))/sigm(idim))*sigm(idim)
      enddo
      do j=1,mbm-1
         l1=j-1
         l2=j
         s=std*(dble(l3-l2)/dble(l3-l1))**0.5d0
         do idim=1,dim
            d=xnew(idim,l3)-xnew(idim,l1)
            d=d-dnint(d/sigm(idim))*sigm(idim)
            xnew(idim,j)=xnew(idim,l1)+d/dble(l3-l1)+xi(s)
            xnew(idim,j)=xnew(idim,j)-
     :              dnint(xnew(idim,j)/sigm(idim))*sigm(idim)
         enddo
      enddo
      return
      end


      function xi(std)
!     sample a gaussian with standard deviation std (box-muller method)
      implicit none
      real*8 xi,std,pi,ranf
      data pi/3.14159265358979323846264338328d0/
      xi=cos(pi*ranf(0.d0))*std*sqrt(-2.d0*log(tiny(pi)+ranf(0.d0)))
      return
      end


      SUBROUTINE DISTR(NN,NORM)
      IMPLICIT NONE
C     WRITES ON FORT.10 THE X-POSITION DISTRIBUTION
      INCLUDE       'mc-sph.par'

      REAL*8        DS,DIST(0:1000)
      INTEGER*8     NN,NORM,I,J,K
      SAVE          DIST

      DS=SIGMA/NN

      DO I=0,NN
         DO J=1,FTN0
            DO K=1,NP
```

```
                IF(-SIGMA/2+(I-.5)*DS.LT.RX(1,J,K).AND.
     :               RX(1,J,K).LT.-SIGMA/2+(I+.5)*DS) THEN
                  DIST(I)=DIST(I)+1.D0
                ENDIF
              ENDDO
            ENDDO
            WRITE(10,*) -SIGMA/2+I*DS,DIST(I)/NORM
          ENDDO

          CLOSE(UNIT=10)

          RETURN
          END


          FUNCTION FACT ( N )
          IMPLICIT NONE
C     FACTORIAL FUNCTION
          INTEGER*8 FACT,N,P,I
          P=1
          DO I=1,N
             P=P*I
          ENDDO
          FACT=P
          END


          SUBROUTINE SUMUP (POTK, OVRLAP, KE, V, W)
          IMPLICIT NONE
C    ***************************************************************
C    ** CALCULATES THE TOTAL ENERGY                                **
C    **                                                            **
C    ** USAGE:                                                     **
C    **                                                            **
C    ** THE SUBROUTINE RETURNS THE TOTAL ENERGY AT THE             **
C    ** BEGINNING AND END OF THE RUN.                              **
C    ***************************************************************
          INCLUDE       'mc-sph.par'

          REAL*8        V, KE, VV, KK
          LOGICAL       OVRLAP
          CHARACTER     POTK*(*)

          REAL*8        RXII, RXIJ, RYIJ, RZIJ
          REAL*8        VIJ, WIJ, RIJSQ, W, WW

          INTEGER*8     TAU, I, J, IDIM

C     POTENTIAL ACTION

          VV      = 0.0
          WW      = 0.0

C    ** LOOP OVER ALL THE PAIRS IN THE LIQUID **

          DO TAU = 1, FTNO
            DO 100 I = 1, NP - 1
               DO 99 J = I + 1, NP
C                 RIJSQ = 0.d0
C                 DO IDIM = 1, DIM
C                    RXIJ = RX(IDIM,TAU,I) - RX(IDIM,TAU,J)
C     ** MINIMUM IMAGE THE PAIR SEPARATIONS **
C                    RXIJ = RXIJ -
C     :                   DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                    RIJSQ = RIJSQ + RXIJ * RXIJ
C                 ENDDO

C     ** EUCLIDEAN DISTANCE
                  RXIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
     :                COS(RX(2,TAU,I))*COS(RX(2,TAU,J))
                  RYIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
     :                SIN(RX(2,TAU,I))*SIN(RX(2,TAU,J))
                  RZIJ  = SIN(RX(1,TAU,I))*SIN(RX(1,TAU,J))
                  RIJSQ = RXIJ + RYIJ + RZIJ
                  RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

                  CALL POT (RIJSQ, VIJ, WIJ, POTK)
                  VV   = VV + VIJ
C                  WW   = WW + WIJ
 99             CONTINUE
                VV = VV - LOG(ABS(RADIUS**2*COS(RX(1,TAU,I))))/LS
                WW = WW - LOG(ABS(RADIUS**2*COS(RX(1,TAU,I))))/LS
 100         CONTINUE
             VV = VV - LOG(ABS(RADIUS**2*COS(RX(1,TAU,NP))))/LS
             WW = WW - LOG(ABS(RADIUS**2*COS(RX(1,TAU,NP))))/LS
          ENDDO
          V = VV
          W = WW


C     KINETIC ACTION

          KK     = 0.0

          DO TAU = 1, FTNO
            DO I = 1, NP
               RXII = RX(1,TAU,I)-RX(1,TAU+1,I)
               RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
               KK=KK+FTM*(RADIUS*RXII)**2./(2.D0*LS**2.)

               RXII = RX(2,TAU,I)-RX(2,TAU+1,I)
               RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
               KK=KK+FTM*(RADIUS*COS(RX(1,TAU,I))*RXII)**2./(2.D0*LS**2.)
            ENDDO
          ENDDO
          KE = KK

          RETURN
          END


          SUBROUTINE SUMUPKE ( KE )
          IMPLICIT NONE
C    ***************************************************************
C    ** CALCULATES THE TOTAL ENERGY                                **
C    **                                                            **
C    ** USAGE:                                                     **
C    **                                                            **
C    ** THE SUBROUTINE RETURNS THE TOTAL ENERGY AT THE             **
C    ** BEGINNING AND END OF THE RUN.                              **
C    ***************************************************************
          INCLUDE       'mc-sph.par'

          REAL*8        KE, KK, RXII
          INTEGER*8     TAU, I

C     KINETIC ACTION

          KK     = 0.0

          DO TAU = 1, FTNO
            DO I = 1, NP
               RXII = RX(1,TAU,I)-RX(1,TAU+1,I)
               RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
               KK=KK+FTM*(RADIUS*RXII)**2./(2.D0*LS**2.)

               RXII = RX(2,TAU,I)-RX(2,TAU+1,I)
               RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
               KK=KK+FTM*(RADIUS*COS(RX(1,TAU,I))*RXII)**2./(2.D0*LS**2.)
            ENDDO
          ENDDO
          KE = KK

          RETURN
          END


          SUBROUTINE PENERGY ( POTK, RXI, I, TAU,
     :      V, W )
          IMPLICIT NONE
C    ***************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOM I WITH ALL OTHER ATOMS.  **
C    **                                                            **
C    ** PRINCIPAL VARIABLES:                                       **
C    **                                                            **
C    ** INTEGER I                 THE ATOM OF INTEREST             **
C    ** INTEGER NP                THE NUMBER OF ATOMS              **
C    ** INTEGER TAU               THE TIMESTEP                     **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS             **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I      **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I           **
C    **                                                            **
C    ** USAGE:                                                     **
C    **                                                            **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY  **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND     **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                        **
C    ***************************************************************
          INCLUDE     'mc-sph.par'

          REAL*8        RXI(MDIM), V, W
          INTEGER*8     I, J, TAU, IDIM
          CHARACTER     POTK*(*)

          REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, VIJ, WIJ

          V      = 0.0
          W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I  **

          DO J = 1, NP

             IF ( I .NE. J ) THEN
                RIJSQ = 0.d0
C                DO IDIM = 1, DIM
C                   RXIJ = RXI(IDIM) - RX(IDIM,TAU,J)

C                   RXIJ = RXIJ -
C     :                DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                   RIJSQ = RIJSQ + RXIJ * RXIJ
C                ENDDO

                RXIJ  = COS(RXI(1))*COS(RX(1,TAU,J))*
     :              COS(RXI(2))*COS(RX(2,TAU,J))
                RYIJ  = COS(RXI(1))*COS(RX(1,TAU,J))*
     :              SIN(RXI(2))*SIN(RX(2,TAU,J))
                RZIJ  = SIN(RXI(1))*SIN(RX(1,TAU,J))
                RIJSQ = RXIJ + RYIJ + RZIJ
                RIJSQ = RADIUS**2*(2*(1.-RIJSQ))
                CALL POT (RIJSQ, VIJ, WIJ, POTK)
                V   = V + VIJ
C                W   = W + WIJ
             ENDIF

          ENDDO

          V = V - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS
          W = W - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS

          RETURN
          END


          SUBROUTINE PPPNERGY ( POTK, RXI, I, RXJ, J,
     :      TAU, V, W )
          IMPLICIT NONE
C    ***************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOMS I AND J WITH         **
C    ** ALL OTHER ATOMS PLUS THE ONE BETWEEN I AND J               **
C    **                                                            **
```

```
C    ** PRINCIPAL VARIABLES:                                          **
C    **                                                               **
C    ** INTEGER I, J              THE ATOMS OF INTEREST                **
C    ** INTEGER NP                THE NUMBER OF ATOMS                  **
C    ** INTEGER TAU               THE TIMESTEP                         **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS                 **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I          **
C    ** REAL*8    RXJ,RYJ,RZJ       THE COORDINATES OF ATOM J          **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I     **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I               **
C    **                                                               **
C    ** USAGE:                                                        **
C    **                                                               **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY      **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND         **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                            **
C    *******************************************************************
        INCLUDE     'mc-sph.par'

        REAL*8        RXI(MDIM), RXJ(MDIM), V, W
        INTEGER*8     I, J, K, TAU, IDIM
        CHARACTER     POTK*(*)

        REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, VIJ, WIJ

        V      = 0.0
        W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I AND J **

        DO 100 K = 1, NP
           IF ( K .NE. I .AND. K .NE. J ) THEN
              RIJSQ = 0.d0
C              DO IDIM = 1, DIM
C                 RXI = RXI(IDIM) - RX(IDIM,TAU,K)

C                 RXIJ = RXIJ -
C     :                DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                 RIJSQ = RIJSQ + RXIJ * RXIJ
C              ENDDO

C     ** EUCLIDEAN DISTANCE
              RXIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :             COS(RXI(2))*COS(RX(2,TAU,K))
              RYIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :             SIN(RXI(2))*SIN(RX(2,TAU,K))
              RZIJ  = SIN(RXI(1))*SIN(RX(1,TAU,K))
              RIJSQ = RXIJ + RYIJ + RZIJ
              RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

              CALL POT (RIJSQ, VIJ, WIJ, POTK)
              V   = V + VIJ
C              W   = W + WIJ

              IF ( I .NE. J ) THEN
                 RIJSQ = 0.d0
C                 DO IDIM = 1, DIM
C                    RXJ = RXJ(IDIM) - RX(IDIM,TAU,K)

C                    RXIJ = RXIJ -
C                    RIJSQ = RIJSQ + RXIJ * RXIJ
C                 ENDDO

C     ** EUCLIDEAN DISTANCE
                 RXIJ  = COS(RXJ(1))*COS(RX(1,TAU,K))*
     :                COS(RXJ(2))*COS(RX(2,TAU,K))
                 RYIJ  = COS(RXJ(1))*COS(RX(1,TAU,K))*
     :                SIN(RXJ(2))*SIN(RX(2,TAU,K))
                 RZIJ  = SIN(RXJ(1))*SIN(RX(1,TAU,K))
                 RIJSQ = RXIJ + RYIJ + RZIJ
                 RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

                 CALL POT (RIJSQ, VIJ, WIJ, POTK)
                 V   = V + VIJ
C                 W   = W + WIJ

              ENDIF

           ENDIF

 100    CONTINUE

C       RETURN

        V = V - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS
        W = W - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS

        IF (I .NE. J) THEN
           RIJSQ = 0.d0
C          DO IDIM = 1, DIM
C             RXI = RXI(IDIM) - RXJ(IDIM)

C             RXIJ = RXIJ -
C     :            DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C             RIJSQ = RIJSQ + RXIJ * RXIJ
C          ENDDO

C     ** EUCLIDEAN DISTANCE
           RXIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          COS(RXI(2))*COS(RXJ(2))
           RYIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          SIN(RXI(2))*SIN(RXJ(2))
           RZIJ  = SIN(RXI(1))*SIN(RXJ(1))
           RIJSQ = RXIJ + RYIJ + RZIJ
           RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

           CALL POT (RIJSQ, VIJ, WIJ, POTK)
           V   = V + VIJ
C           W   = W + WIJ
           V = V - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
           W = W - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
        ENDIF

        RETURN
        END


        SUBROUTINE PPNERGY ( POTK, RPI, I, RPJ, J,
     :     TAU, V, W )
        IMPLICIT NONE
C    *******************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOMS I AND J WITH             **
C    ** ALL OTHER ATOMS PLUS THE ONE BETWEEN I AND J                   **
C    **                                                               **
C    ** PRINCIPAL VARIABLES:                                          **
C    **                                                               **
C    ** INTEGER I, J              THE ATOMS OF INTEREST                **
C    ** INTEGER NP                THE NUMBER OF ATOMS                  **
C    ** INTEGER TAU               THE TIMESTEP                         **
C    ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS                 **
C    ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I          **
C    ** REAL*8    RXJ,RYJ,RZJ       THE COORDINATES OF ATOM J          **
C    ** REAL*8    V                 THE POTENTIAL ENERGY OF ATOM I     **
C    ** REAL*8    W                 THE VIRIAL OF ATOM I               **
C    **                                                               **
C    ** USAGE:                                                        **
C    **                                                               **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY      **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND         **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                            **
C    *******************************************************************
        INCLUDE     'mc-sph.par'

        REAL*8        RPI(MDIM),RPJ(MDIM),RXI(MDIM),RXJ(MDIM), V, W
        INTEGER*8     I, J, K, TAU, IDIM
        CHARACTER     POTK*(*)

        REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, VIJ, WIJ

C     MAPPING
        RXI(1)=RPI(1)
        RXI(2)=RPI(2)
        RXJ(1)=RPJ(1)
        RXJ(2)=RPJ(2)

        V      = 0.0
        W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I AND J **

        DO 100 K = 1, NP
           IF ( K .NE. I .AND. K .NE. J ) THEN
              RIJSQ = 0.d0
C              DO IDIM = 1, DIM
C                 RXI = RXI(IDIM) - RX(IDIM,TAU,K)

C                 RXIJ = RXIJ -
C     :                DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                 RIJSQ = RIJSQ + RXIJ * RXIJ
C              ENDDO

C     ** EUCLIDEAN DISTANCE
              RXIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :             COS(RXI(2))*COS(RX(2,TAU,K))
              RYIJ  = COS(RXI(1))*COS(RX(1,TAU,K))*
     :             SIN(RXI(2))*SIN(RX(2,TAU,K))
              RZIJ  = SIN(RXI(1))*SIN(RX(1,TAU,K))
              RIJSQ = RXIJ + RYIJ + RZIJ
              RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

              CALL POT (RIJSQ, VIJ, WIJ, POTK)
              V   = V + VIJ
C              W   = W + WIJ

              IF ( I .NE. J ) THEN
                 RIJSQ = 0.d0
C                 DO IDIM = 1, DIM
C                    RXJ = RXJ(IDIM) - RX(IDIM,TAU,K)

C                    RXIJ = RXIJ -
C     :                DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C                    RIJSQ = RIJSQ + RXIJ * RXIJ
C                 ENDDO

C     ** EUCLIDEAN DISTANCE
                 RXIJ  = COS(RXJ(1))*COS(RX(1,TAU,K))*
     :                COS(RXJ(2))*COS(RX(2,TAU,K))
                 RYIJ  = COS(RXJ(1))*COS(RX(1,TAU,K))*
     :                SIN(RXJ(2))*SIN(RX(2,TAU,K))
                 RZIJ  = SIN(RXJ(1))*SIN(RX(1,TAU,K))
                 RIJSQ = RXIJ + RYIJ + RZIJ
                 RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

                 CALL POT (RIJSQ, VIJ, WIJ, POTK)
                 V   = V + VIJ
C                 W   = W + WIJ

              ENDIF

           ENDIF

 100    CONTINUE

C       RETURN

        V = V - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS
        W = W - LOG(ABS(RADIUS**2*COS(RXI(1))))/LS

        IF (I .NE. J) THEN
           RIJSQ = 0.d0
C          DO IDIM = 1, DIM
```

```
C             RXIJ = RXI(IDIM) - RXJ(IDIM)

C             RXIJ = RXIJ -
C     :            DNINT ( RXIJ/SIGM(IDIM) )*SIGM(IDIM)
C             RIJSQ = RIJSQ + RXIJ * RXIJ
C           ENDDO

C     ** EUCLIDEAN DISTANCE
            RXIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          COS(RXI(2))*COS(RXJ(2))
            RYIJ  = COS(RXI(1))*COS(RXJ(1))*
     :          SIN(RXI(2))*SIN(RXJ(2))
            RZIJ  = SIN(RXI(1))*SIN(RXJ(1))
            RIJSQ = RXIJ + RYIJ + RZIJ
            RIJSQ = RADIUS**2*(2*(1.-RIJSQ))

            CALL POT (RIJSQ, VIJ, WIJ, POTK)
            V   = V + VIJ
C            W   = W + WIJ
            V = V - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
            W = W - LOG(ABS(RADIUS**2*COS(RXJ(1))))/LS
          ENDIF

          RETURN
          END


          SUBROUTINE KENERGY ( RXI, I, TAU, KE )
          IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C     **                                                              **
C     ** PRINCIPAL VARIABLES:                                         **
C     **                                                              **
C     ** INTEGER I                 THE ATOM OF INTEREST               **
C     ** INTEGER NP                THE NUMBER OF ATOMS                **
C     ** INTEGER TAU               THE TIMESTEP                       **
C     ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS               **
C     ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I        **
C     ** REAL*8    KE                THE KINETIC ENERGY OF ATOM I     **
C     **                                                              **
C     ** USAGE:                                                       **
C     **                                                              **
C     ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C     ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C     ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C     ****************************************************************
          INCLUDE     'mc-sph.par'

          REAL*8        RXI(MDIM), KE
          REAL*8        RXIP1, RXIS1, RXIP2, RXIS2, RXII
          INTEGER*8     I, TAU, IDIM

          KE = 0.d0

          RXIP1 = RX(1,TAU-1,I)
          RXIS1 = RX(1,TAU+1,I)
          RXII = RXI(1) - RXIP1
          RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
          KE=KE+0.5*FTM*(RADIUS*RXII/LS)**2.
          RXII = RXI(1) - RXIS1
          RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
          KE=KE+0.5*FTM*(RADIUS*RXII/LS)**2.

          RXIP2 = RX(2,TAU-1,I)
          RXIS2 = RX(2,TAU+1,I)
          RXII = RXI(2) - RXIP2
          RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
          KE=KE+0.5*FTM*(RADIUS*COS(RXIP1)*RXII/LS)**2.
          RXII = RXI(2) - RXIS2
          RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
          KE=KE+0.5*FTM*(RADIUS*COS(RXI(1))*RXII/LS)**2.

          RETURN
          END


          SUBROUTINE KKNERGY ( RXI, IP, TAU, KE )
          IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C     **                                                              **
C     ** PRINCIPAL VARIABLES:                                         **
C     **                                                              **
C     ** INTEGER I                 THE ATOM OF INTEREST               **
C     ** INTEGER NP                THE NUMBER OF ATOMS                **
C     ** INTEGER TAU               THE TIMESTEP                       **
C     ** REAL*8    RX, RY, RZ        THE ATOM POSITIONS               **
C     ** REAL*8    RXI,RYI,RZI       THE COORDINATES OF ATOM I        **
C     ** REAL*8    KE                THE KINETIC ENERGY OF ATOM I     **
C     **                                                              **
C     ** USAGE:                                                       **
C     **                                                              **
C     ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C     ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C     ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C     ****************************************************************
          INCLUDE     'mc-sph.par'

          REAL*8        RXI(MDIM), KE
          REAL*8        RXIS, RXII
          INTEGER*8     IP, TAU, IDIM

          KE = 0.d0

          RXIS = RX(1,TAU+1,IP)
          RXII = RXI(1) - RXIS
          RXII = RXII - DNINT ( RXII/SIGM(1) )*SIGM(1)
          KE=KE+0.5*FTM*(RADIUS*RXII/LS)**2.

          RXIS = RX(2,TAU+1,IP)
          RXII = RXI(2) - RXIS
          RXII = RXII - DNINT ( RXII/SIGM(2) )*SIGM(2)
          KE=KE+0.5*FTM*(RADIUS*COS(RXI(1))*RXII/LS)**2.

          RETURN
          END


          SUBROUTINE CWIND ( WIND )
          IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS THE WINDING NUMBER.                                  **
C     ****************************************************************
          INCLUDE     'mc-sph.par'

          REAL*8        WIND, WI, SURF
          REAL*8        RX11, RX21, RX31, RX12, RX22, RX32
          INTEGER*8     I, J, IDIM

          WIND = 0.d0
          SURF = 4*PI*RADIUS**2
          DO I = 1, NP
             DO J = 1, FTN0 - 1
                RX11 = RX(1,J  ,I)
                RX21 = RX(1,J+1,I)
                RX31 = RX(1,J+2,I)
                RX12 = RX(2,J  ,I)
                RX22 = RX(2,J+1,I)
                RX32 = RX(2,J+2,I)
                WI = ((RX21-RX11)*(RX32-RX22)-(RX22-RX12)*(RX31-RX21))*
     :             RADIUS**2*COS(RX21)
                WIND = WIND + WI
             ENDDO
          ENDDO
          IF (WIND.GT.SURF) WIND = SURF
          WIND = WIND**2

 20       CONTINUE
          RETURN
          END


          REAL*8 FUNCTION RANF ( DUMMY )
          IMPLICIT NONE
C     ****************************************************************
C     ** RETURNS A UNIFORM RANDOM VARIATE IN THE RANGE 0 TO 1.         **
C     **                                                              **
C     **                 ***************                              **
C     **                 **  WARNING  **                              **
C     **                 ***************                              **
C     **                                                              **
C     ** GOOD RANDOM NUMBER GENERATORS ARE MACHINE SPECIFIC.           **
C     ** PLEASE USE THE ONE RECOMMENDED FOR YOUR MACHINE.              **
C     **                                                              **
C     ** RAND(FLAG) returns a pseudo-random number from a uniform      **
C     ** distribution between 0 and 1. If FLAG is 0, the next number   **
C     ** in the current sequence is returned; if FLAG is 1, the        **
C     ** generator is restarted by CALL SRAND(0); if FLAG has any      **
C     ** other value, it is used as a new seed with SRAND.             **
C     ****************************************************************
          INTEGER*8   L, C, M
          PARAMETER ( L = 1029, C = 221591, M = 1048576 )
          INTEGER*8   SEED
          SAVE        SEED
          DATA        SEED / 0 /
          REAL*8      DUMMY

C          SEED = MOD ( SEED * L + C, M )
C          RANF = DBLE( SEED ) / M

          RANF = RAND ( )

          RETURN
          END


          SUBROUTINE READCN ( CNFILE )
          IMPLICIT NONE
C     ****************************************************************
C     ** SUBROUTINE TO READ IN THE CONFIGURATION FROM UNIT 10          **
C     ****************************************************************
          INCLUDE     'mc-sph.par'

          CHARACTER   CNFILE*(*), HASH*1, MYFMT*77

          INTEGER*8   CNUNIT, I, J, NNP, NI, IDIM
          PARAMETER ( CNUNIT = 10 )

C    ****************************************************************

          OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'OLD' )
          WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X))'

          READ ( CNUNIT,* ) HASH, FTN0, NNP
          IF ( NNP .NE. NP ) STOP 'N ERROR IN READCN'

          DO 100 I = 1, NNP
             READ ( CNUNIT,* ) HASH, NEXT(I)
             NI = NEXT(I)
             DO 90 J = 1, FTN0+1
                READ ( CNUNIT, MYFMT) LEXT(I), (RX(IDIM,J,I),IDIM=1,DIM)
 90          ENDDO
             READ ( CNUNIT,13 )
             READ ( CNUNIT,13 )
             DO IDIM = 1, DIM
                RX(IDIM,0,NI) = RX(IDIM,FTN0,I)
             ENDDO
 100      ENDDO
 13       FORMAT(A2)

          CLOSE ( UNIT = CNUNIT )
```

```
      RETURN
      END


      SUBROUTINE WRITCN ( CNFILE )
      IMPLICIT NONE
C    ****************************************************************
C    ** SUBROUTINE TO WRITE OUT THE CONFIGURATION TO UNIT 10         **
C    ****************************************************************
      INCLUDE     'mc-sph.par'

      CHARACTER   CNFILE*(*), MYFMT*77

      INTEGER*8   CNUNIT, I, J, IDIM
      REAL*8      R
      PARAMETER ( CNUNIT = 10 )

C   ****************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X))'

      WRITE(*,*) 'output to file -----------------------------------
    :---------------'
      CALL FLUSH ( CNUNIT )
      WRITE ( CNUNIT,* ) '#',FTNO,NP

      DO I = 1, NP
         WRITE ( CNUNIT,* ) '#', NEXT(I)
         DO J = 1, FTNO+1
C            WRITE ( CNUNIT, MYFMT ) LEXT(I),
C     :    (RX(IDIM,J,I)-DNINT(RX(IDIM,J,I)/SIGMA)*SIGMA,IDIM=1,DIM)
            WRITE ( CNUNIT, MYFMT ) LEXT(I),
    :             (RX(IDIM,J,I),IDIM=1,DIM)
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE INITCNSPH ( CNFILE )
      IMPLICIT NONE
C    ****************************************************************
C    ** SUBROUTINE TO INITIALIZE THE CONFIGURATION TO UNIT 10        **
C    ****************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8      RANF
      CHARACTER   CNFILE*(*), MYFMT*77

      INTEGER*8   CNUNIT, I, J, IDIM
      REAL*8      RXI(DIM)
      PARAMETER ( CNUNIT = 10 )

C   ****************************************************************

      OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
      WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X))'

      WRITE ( CNUNIT,* ) '#',FTNO,NP

      DO I = 1, NP
         WRITE ( CNUNIT,* ) '#', NEXT(I)
         RXI(1) = 0.
         RXI(2) = SIGM(2)*(RANF(0.d0)-.5)
         DO J = 1, FTNO+1
            WRITE ( CNUNIT, MYFMT ) I, RXI(1), RXI(2)
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE POT (RIJSQ, VIJ, WIJ, POTK)
      IMPLICIT NONE
C    ****************************************************************
C    ** SUBROUTINE FOR THE PAIR POTENTIAL                           **
C    ****************************************************************
      INCLUDE     'mc-sph.par'

      CHARACTER POTK*(*)
      REAL*8    RIJ, RIJSQ, RMIN, RMSQ, RCSQ, SR2, SR6, VIJ, WIJ
      REAL*8    COU3, QHS3, F
      PARAMETER ( COU3 = 4.76015472795910701328763470057d0 )
      PARAMETER ( QHS3 = 24.d0 )

C       PI = ACOS(-1.d0)

      VIJ = 0.d0
      WIJ = 0.d0
      IF (DIM .GT. 3) THEN
         WRITE(*,*) "DIM > 3 !!!"
         STOP
      ENDIF
      IF (POTK .EQ. 'FREE') RETURN

      IF (POTK .EQ. 'LJ') THEN
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C     W. L. McMillan, Phys. Rev. A 138, 442 (1965)     C
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C          RMIN = 2.d0**(1./6)
C          RMIN = EPSA
C          RMIN = TINY(PI)
         RMIN = 0.d0
         RIJSQ = RIJSQ / (SIG * SIG)
         RMSQ = RMIN*RMIN
         RCSQ = RCUT*RCUT
         IF (RIJSQ .LT. RMSQ) THEN
            RIJSQ = RMSQ
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ELSEIF (RIJSQ .GT. RMSQ .AND. RIJSQ .LT. RCSQ) THEN
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ENDIF
      ELSEIF (POTK .EQ. 'BUMP') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG) THEN
            VIJ = EPSR
         ENDIF
      ELSEIF (POTK .EQ. 'PSW') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG) THEN
            VIJ = EPSR
         ELSEIF (RIJ .LE. SIG + RCUT .AND. RIJ .GT. SIG) THEN
            VIJ = -EPSA
         ENDIF
      ELSEIF (POTK .EQ. 'COULOMB') THEN
         F = NP/(NP-1)
         RIJ = SQRT( RIJSQ )
         IF (RIJ .GT. TINY(PI)) THEN
            IF (DIM .EQ. 3) THEN
               VIJ = -F*COU3*SIG/SIGMA/2.
               VIJ = VIJ + SIG/RIJ
            ELSEIF (DIM .EQ. 2) THEN
               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/SIG))/4.
               VIJ = VIJ - LOG(RIJ/SIG)
            ELSEIF (DIM .EQ. 1) THEN
               VIJ = F*SIGMA/SIG/4.
               VIJ = VIJ - RIJ/SIG
            ENDIF
         ELSE
            RIJ = TINY(PI)
            IF (DIM .EQ. 3) THEN
               VIJ = -F*COU3*SIG/SIGMA/2.
               VIJ = VIJ + SIG/RIJ
            ELSEIF (DIM .EQ. 2) THEN
               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/SIG))/4.
               VIJ = VIJ - LOG(RIJ/SIG)
            ELSEIF (DIM .EQ. 1) THEN
               VIJ = F*SIGMA/SIG/4.
               VIJ = VIJ - RIJ/SIG
            ENDIF
         ENDIF
      ELSEIF (POTK .EQ. 'QHS') THEN
         F = NP/(NP-1)
         IF (DIM .EQ. 3) THEN
            IF (RIJSQ .LE. SIG*SIG) THEN
               VIJ = 10.**10.
            ELSE
            VIJ = -F*QHS3*(SIGMA-SIG)/SIGMA**3/2./FTM
            VIJ = VIJ + (2.*RIJSQ+SIG**2.)/(RIJSQ-SIG*SIG)**2./2./FTM
            WIJ = -2.*RIJSQ*(RIJSQ+2.*SIG*SIG)/(RIJSQ-SIG*SIG)**3./FTM
            ENDIF
         ELSE
            WRITE(*,*) 'QHS in dimension different from 3'
            STOP
         ENDIF
      ELSEIF (POTK .EQ. 'HARMONIC') THEN
         VIJ = 0.5*RIJSQ/SIG**2.
      ENDIF

      RETURN
      END


      SUBROUTINE GOFR ( TAU, MAXBIN, HIST )
      IMPLICIT NONE
C    ****************************************************************
C    ** CALCULATES THE RADIAL DISTRIBUTION FUNCTION HISTOGRAM.      **
C    ****************************************************************
      INCLUDE     'mc-sph.par'

      REAL*8        RXIJ, RYIJ, RZIJ, RIJSQ, RIJ
      INTEGER*8     TAU, IDIM, I, J, BIN, MAXBIN, HIST(MAXBIN)

      DO I = 1, NP - 1
         DO J = I + 1, NP
C     CALCULATE HISTOGRAM FOR g(r)

C     Euclidean Distance
            RXIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
    :             COS(RX(2,TAU,I))*COS(RX(2,TAU,J))
            RYIJ  = COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*
    :             SIN(RX(2,TAU,I))*SIN(RX(2,TAU,J))
            RZIJ  = SIN(RX(1,TAU,I))*SIN(RX(1,TAU,J))
            RIJSQ = RXIJ + RYIJ + RZIJ
c            RIJ = 2*RADIUS*SIN(ACOS(RIJSQ)/2.)
            RIJ = RADIUS*SQRT(2.-2*RIJSQ)

C     Geodesic Distance
```

```
c             RIJ = RADIUS*ACOS(SIN(RX(1,TAU,I))*SIN(RX(1,TAU,J))+
c     : COS(RX(1,TAU,I))*COS(RX(1,TAU,J))*COS(RX(2,TAU,I)-RX(2,TAU,J)))

              BIN = INT ( RIJ/DELR ) + 1

              IF (BIN .LE. MAXBIN) THEN
                 HIST(BIN) = HIST(BIN) + 2
              ENDIF

           ENDDO
        ENDDO

        RETURN
        END


        SUBROUTINE MAPPINGD()
        IMPLICIT NONE
C    ****************************************************************
C    ** CHANGE COORDINATES (DIRECT)                                **
C    ****************************************************************
        INCLUDE     'mc-sph.par'

        INTEGER*8 I,J

        DO I=1,NP
           DO J=0,FTN0+1
              RP(1,J,I)=RX(1,J,I)
              RP(2,J,I)=RX(2,J,I)
           ENDDO
        ENDDO

        RETURN
        END


        SUBROUTINE MAPPINGI()
        IMPLICIT NONE
C    ****************************************************************
C    ** CHANGE COORDINATES (INVERSE)                               **
C    ****************************************************************
        INCLUDE     'mc-sph.par'

        INTEGER*8 I,J

        DO I=1,NP
           DO J=0,FTN0+1
              RX(1,J,I)=RP(1,J,I)
              RX(2,J,I)=RP(2,J,I)
           ENDDO
        ENDDO

        RETURN
        END


        SUBROUTINE GAUSS(CP,GA)
        IMPLICIT NONE
C    ****************************************************************
C    ** GAUSSIAN TRANSITION PROBABILITY FOR BROWNIAN BRIDGE        **
C    ****************************************************************
        INCLUDE     'mc-sph.par'

        INTEGER*8 I,J,K,CP,MCM
        REAL*8 GA,VAR,DD

        MCM = CP+MBM-FLOOR((CP+MBM-.1)/FTN0)*FTN0
        VAR = 2*LS/FTM

        GA = 0.D0
        DO I=1,DIM
           DO J=1,NP
              IF(CP+MBM.LE.FTN0)THEN
                 DO K=CP+1,MCM
                    DD=RPN(I,K,J)-RPN(I,K-1,J)
                    DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                    GA=GA+DD**2
                    DD=RP(I,K,J)-RP(I,K-1,J)
                    DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                    GA=GA-DD**2
                 ENDDO
              ELSE
                 DO K=CP+1,FTN0
                    DD=RPN(I,K,J)-RPN(I,K-1,J)
                    DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                    GA=GA+DD**2
                    DD=RP(I,K,J)-RP(I,K-1,J)
                    DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                    GA=GA-DD**2
                 ENDDO
                 DO K=1,MCM
                    DD=RPN(I,K,J)-RPN(I,K-1,J)
                    DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                    GA=GA+DD**2
                    DD=RP(I,K,J)-RP(I,K-1,J)
                    DD=DD-DNINT ( DD/SIGM(I) )*SIGM(I)
                    GA=GA-DD**2
                 ENDDO
              ENDIF
           ENDDO
        ENDDO

        GA = EXP(GA/VAR)

        RETURN
        END


        SUBROUTINE EQUIV(A,B)
        IMPLICIT NONE
C    ****************************************************************
C    ** A = B WITH A, B TWO PATHS                                  **
C    ****************************************************************
        INCLUDE     'mc-sph.par'

        INTEGER*8 I,J,K
        REAL*8    A(MDIM,0:N,MNP),B(MDIM,0:N,MNP)

        DO I=1,DIM
           DO J=0,FTN0+1
              DO K=1,NP
                 A(I,J,K)=B(I,J,K)
              ENDDO
           ENDDO
        ENDDO

        RETURN
        END

*****************************************************************************
*** mc-sph.par ***********************************************************
*****************************************************************************

     REAL*8        PI
     REAL*8        DELR
        INTEGER*8     MDIM, N, MNP

        PARAMETER ( PI = 3.14159265358979323846264338328d0 )

     PARAMETER ( MDIM = 10   )   ! maximum number of dimensions
     PARAMETER ( MNP  = 1000 )   ! maximum number of particles
     PARAMETER ( N    = 3000 )   ! maximum number of time slices
     PARAMETER ( DELR = 0.01d0 ) ! grid spacing for g(r)

     INTEGER*8     DIM
        REAL*8        RX(MDIM,0:N,MNP)
     REAL*8        RPN(MDIM,0:N,MNP),RP(MDIM,0:N,MNP)
     REAL*8        FTM, LS, SIGMA, SIGM(MDIM), RADIUS
        REAL*8        SIG, EPSA, EPSR, EPS, RCUT
        INTEGER*8     FTN0, NP
     INTEGER*8     NEXT(MNP), LEXT(MNP), MEXT(N,MNP), MBM

        ! path
        COMMON / BLOCK1 / RX, RP, RPN, DIM
        ! pair-potential parameters
        COMMON / BLOCK2 / SIG, EPSR, EPSA, EPS, RCUT
        ! mass, timestep, box edge, # timeslices, # particles, radius
        COMMON / BLOCK3 / FTM, LS, SIGMA, SIGM, FTN0, NP, RADIUS
        ! permutations
        COMMON / BLOCK4 / NEXT, LEXT, MEXT, MBM

*****************************************************************************
*** data-gr-rs.in ********************************************************
*****************************************************************************

0 number of spatial dimensions          (2 sphere)
2
1 seed of the random sequence                 RAND
3
2 if bose                                    (T/F)
T
3 number of particles                     (<= MNP)
10
4 the potential                     SUBROUTINE POT
FREE
5 # time slices = 1/temperature/timestep     (< N)
50
6 mass                             (hbar = kb = 1)
1.d0
7 # of cycles                              (nstep)
500000000000000000
8 # of steps between output lines         (iprint)
100
9 # of steps between configuration saves   (isave)
100
10 # of steps for equilibration            (iequi)
100
11 # of steps for acceptance ratios       (iratio)
100
12 # of steps for rdf calculation    (icalcg < #5)
10
13 configuration file name
conf.xyz
14 0 init zero, 1 init, 2 rest conf, 3 rest full
1
15 radius of the sphere
5.d0
16 temperature                      THERMODYNAMICS
.03d0
17 maximum displacement/box edge
.05d0
18 maximum # of bridge timeslices     (> 1; <= #5)
10
19 potential cutoff distance (LJ)   SUBROUTINE POT
2.d0
20 sig    (LJ 2.566)                SUBROUTINE POT
1.d0
21 epsr   (LJ INIT)                 SUBROUTINE POT
1.d10
22 epsa                             SUBROUTINE POT
1.d0
23 eps    (LJ 10.22)                SUBROUTINE POT
1.d1
24 if only displace                          (T/F)
F
25 if only bridge                            (T/F)
F
```

## AUTHOR DECLARATIONS

### Conflicts of interest

None declared.

### Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

### Funding

None declared.

[1] R. Fantoni, Coherent State Path Integral Monte Carlo, Eur. Phys. J. D (2025), under review.
[2] D. M. Ceperley, Path integrals in the theory of condensed Helium, Rev. Mod. Phys. **67**, 279 (1995).
[3] D. M. Ceperley, Fermion Nodes, J. Stat. Phys. **63**, 1237 (1991).
[4] D. M. Ceperley, Path integral Monte Carlo methods for fermions, in *Monte Carlo and Molecular Dynamics of Condensed Matter Systems*, edited by K. Binder and G. Ciccotti (Editrice Compositori, Bologna, Italy, 1996).